Question			Answer	Marks	AO	Guidance
6	(a)		$\sin x\cos\frac{\pi}{6}+\cos x\sin\frac{\pi}{6}$ $=\cos x\cos\frac{\pi}{4}+\sin x\sin\frac{\pi}{4}$	M1	2.1	Using a compound angle formula at least once
			$\frac{\sqrt{3}}{2}\sin x+\frac{1}{2}\cos x=\frac{\sqrt{2}}{2}\cos x+\frac{\sqrt{2}}{2}\sin x$	M1	2.1	Uses exact values for at least 2 trigonometric terms
			$\sin x\left(\frac{\sqrt{3}}{2}-\frac{\sqrt{2}}{2}\right)=\cos x\left(\frac{\sqrt{2}}{2}-\frac{1}{2}\right)$	M1	2.1	Collecting terms and factorising
			$\tan x=\frac{\sqrt{2}-1}{\sqrt{3}-\sqrt{2}}$	E1	2.1	**AG** Complete argument with proper use of brackets where necessary
				[4]		
6	(b)		$x=\frac{7\pi}{24},\frac{31\pi}{24}$	B1	1.1b	Allow for both values without working and no others in the range $0\le x\le 2\pi$. Allow decimal equivalents 0.916, 4.06 or better
				[1]		